
A MULTIMODAL APPROACH TO DERMATOLOGY VQA

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ABSTRACT

Artificial Intelligence has been making significant progress across various fields, not limited only to communication and text generation tasks, but also in health-care; particularly in areas such as dermatology. Solutions such as image-based analysis have helped improve diagnostic accuracy. However, the integration of interactive systems capable of visual question answering (VQA) remains largely unexplored. This project aspires to extend the functionality of PanDerm(Yan et al. (2024)), a dermatology-focused foundation model, by fine-tuning a language model on top of it resulting in a multi-model able to address VQA tasks. In the absence of dedicated VQA datasets for dermatology, we propose augmenting the ISIC 2018(Codella et al. (2018) and Tschandl et al. (2018)) dataset to create a VQA-compatible dataset. This augmentation enables the generation of tailored data, allowing PanDerm to learn associations between images, their diagnoses, and corresponding question-answer pairs. The project aims to accurately detect skin conditions and provide expert-level, interactive responses to user queries, enabling personalized and accessible dermatological care.

1 INTRODUCTION

This project aims to fine-tune large language models after fusing them with the PanDerm foundation model (Yan et al. (2024)) to perform VQA tasks. To address the lack of dedicated VQA datasets in dermatology, we integrated two complementary datasets: the ISIC 2019 dataset, which contains labeled skin images, and an information-seeking question-answer dataset. We constructed a training dataset by aligning the diagnostic labels from ISIC images with relevant questions and answers about similar diagnoses.

This project aims to improve PanDerm by enabling it to not only detect skin conditions but also provide answers to questions about symptoms, treatment options, and possible causes. The resulting system aspires to empower clinicians and patients alike by improving accessibility to healthcare and offering expert-level insights in real time. This paper outlines our methodology for dataset alignment, fine-tuning, and evaluation, highlighting the potential of foundation models to transform interactive healthcare solutions.

2 LITERATURE REVIEW

PanDerm is a multimodal foundation model designed for dermatology, leveraging self-supervised learning to process over 2 million skin disease images across four imaging modalities. Its architecture leverages techniques like the drop path regularization method, which enhances model robustness by stochastically dropping connections during training to prevent overfitting and improve

generalization (Huang et al. (2016)). Its also integrates visual features with clinical context, enabling it to handle diverse dermatological tasks, including diagnosis, segmentation, and prognosis. PanDerm’s ability to operate with minimal labeled data while also delivering state-of-the-art results makes it ideal for developing a VQA system tailored for dermatology, addressing the complexity of multimodal inputs and clinical queries (Yan et al. (2024)). Furthermore, we also looked at Gan et al. (2022) to better understand how to fuse vision language models and their architectures, particularly exploring dual encoders, fusion encoders, and cross-modal attention mechanisms to enhance multimodal integration.

2.1 VISUAL QUESTION ANSWERING

Visual Question Answering (VQA) is a prominent area of research that focuses on enabling models to interpret visual inputs alongside textual queries to generate contextually accurate answers. In a typical VQA task, the model is presented with an image and a question and is expected to generate an answer by understanding the visual context of the image.

VQA tasks are commonly approached in two formats: (1) multiple-choice questions, where the model selects the correct option from predefined answers, and (2) open-ended questions, which require descriptive and unconstrained responses.

In recent years, several benchmarks have been developed to advance VQA research. For instance, Visual Dialog (Das et al. (2017)) extends traditional VQA by incorporating multi-turn dialog interactions. Similarly, datasets like TextVQA (Singh et al. (2019)) and ST-VQA (Biten et al. (2019)) focus on extracting textual information embedded within scenes in images.

Despite substantial progress in general VQA tasks, there has been limited exploration of VQA in medical domains like dermatology. Recently, the DermaVQA dataset (Yim et al. (2024)) was introduced as a benchmark for dermatology-specific VQA tasks. Their baseline model combines two components: an image-to-text generator (LLaVA-Med), which diagnoses skin conditions, and a text-to-text module powered by GPT-4 that generates user-friendly responses based on the diagnosis and query. Additionally, efforts such as IKIM at MEDIQA-M3G 2024 (Bauer et al. (2024)) demonstrated the potential to combine open source medical vision language models with large language models for cross-lingual visual question answering.

Building upon these advancements, we leverage PanDerm, a dermatology-focused foundation model (Yan et al. (2024)), to develop a visual-language model tailored for dermatology. Our approach incorporates lightweight language models and employs fusion techniques such as Cross-Attention and UNITER to integrate visual and textual modalities effectively.

3 DATASET

Given the limited availability of Visual Question Answering (VQA) datasets specific to dermatology, we created a tailored dataset to fine-tune PanDerm for VQA tasks. Our work primarily focuses on eight dermatological diagnoses: *Melanoma (MEL)*, *Vascular lesion (VASC)*, *Melanocytic nevus (NV)*, *Actinic keratosis (AK)*, *Benign keratosis (BKL)*, *Basal cell carcinoma (BCC)*, *Dermatofibroma (DF)*, and *Squamous cell carcinoma (SCC)*.

We primarily utilized the ISIC 2018 (Codella et al. (2018) and Tschandl et al. (2018)) dataset, which was also employed by PanDerm. Each image in the ISIC dataset is associated with a corresponding diagnosis. To create a VQA-compatible dataset, we augmented this dataset by generating generic questions corresponding to the diagnosis of each image, such as "What is the diagnosis in the image?" and "What are its causes?". By doing so, the augmented dataset contains $n \times q$ data samples, where n is the number of images in the initial dataset, and q is the number of questions per image.

To construct this dataset, we employed a combination of techniques, sourcing data both from pre-existing datasets and web-based resources.

3.1 AUGMENTING WITH EXISTING DATASETS

For *Melanoma (MEL)* and *Basal Cell Carcinoma (BCC)*, we leveraged a dermatology-based question-answer dataset available on Hugging Face (Mreeb (2024)). Using an iterative process

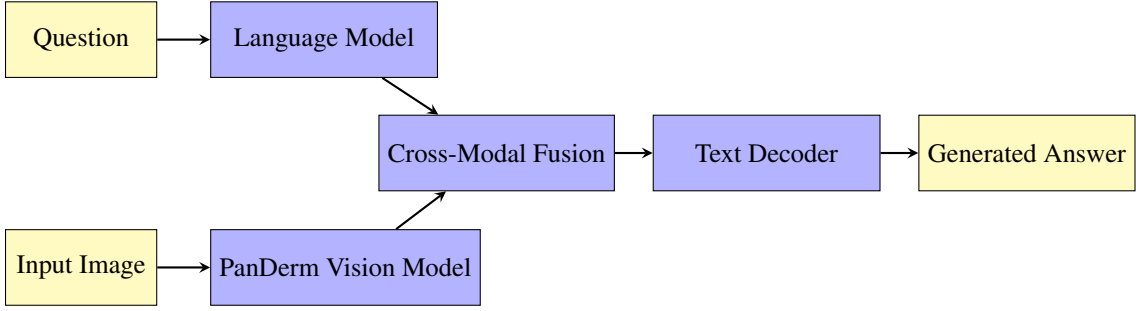


Figure 1: Architecture of the Visual Question Answering (VQA) Model. The model incorporates a visual encoder (PanDerm), a language model (BERT), and a decoder (GPT) with cross-modal attention to generate textual answers.

for each diagnosis, we parsed through the dataset to identify prompts that directly referenced "melanoma" and "basal cell carcinoma," as well as their respective abbreviations (*MEL* and *BCC*). The selected prompts, along with their corresponding answers, were formatted into a JSON structure and integrated into our dataset file.

3.2 AUGMENTING WITH WEB-SOURCED DATA

For the remaining diagnoses (*NV*, *AK*, *BKL*, *DF*, *VASC*, and *SCC*), we manually curated question-answer pairs from various web sources that are considered as a reliable dermatology-related inquiries sources (Anna Hernández (2024) and Altruderm (2024)). Frequently asked questions relevant to each diagnosis were collected and transformed into a VQA-compatible format. These included typical user queries regarding symptoms, treatment options, and diagnostics. The curated data was formatted into the same JSON structure as the entries in the section 3.1 and appended to the dataset.

3.3 ADDITIONAL DATASETS

In addition to the aforementioned datasets, we incorporate DermaVQA, a recently introduced benchmark for dermatology-specific visual question answering (Yim et al., 2024). The dataset is curated from two distinct sources: IYYI, a Chinese medical platform, and Reddit, specifically the subreddit r/DermatologyQuestions.

IYYI: This platform allows users to submit medical questions accompanied by images, which are answered by healthcare professionals. It serves as a structured source of medically reliable content.

Reddit: On the subreddit r/DermatologyQuestions, users post dermatology-related queries and receive answers from the community. Since responses may not always be written by medical professionals, the dataset creators employed certified dermatologists to draft accurate answers for these cases.

Data collection was achieved using web-crawling techniques: crawling for IYYI and PRAW, a Python API, for Reddit. Post-collection, the dataset was filtered to remove images containing identifiable features (e.g., tattoos) or inappropriate content (e.g., images of genitalia). A gold-standard dataset was curated through meticulous post-processing to ensure high-quality, anonymized data for benchmarking.

4 METHODOLOGY

This project implements a multimodal Visual Question Answering (VQA) system by leveraging a cross-modal attention mechanism to combine visual and textual modalities. The system is designed to process input images along with textual queries to generate accurate textual answers.

4.1 ARCHITECTURE

The model uses a Cross-Modal Attention mechanism built using PyTorch’s MultiHeadAttention module. This method helps the system connect textual questions with the relevant parts of an image, enabling it to reason across both text and visuals. Specifically, the query originates from the textual input (question), while the keys and values are derived from the image.

The architecture is composed of the following components:

- **Visual Model:** We use PanDerm, a foundation model pre-trained on dermatological images, as our vision encoder. PanDerm extracts the necessary visual features from skin lesion images which are crucial for accurate reasoning and answering questions about the image.
- **Language Model:** The text input (question) is encoded using BERT (bert-base-uncased), a transformer-based language model. BERT is employed to generate embeddings for the input questions.
- **Decoder:** To produce textual answers, the architecture incorporates a GPT-based decoder. The decoder takes fused multimodal features (from the cross-modal attention) as input and generates natural language answers.

4.2 TRAINING PROCESS

The data preparation process described in Section 3 was used to construct a multimodal dataset consisting of images, associated diagnoses, questions, and answers. The dataset was further split into training, validation, and test subsets. PyTorch’s DataLoader was employed to manage batching, shuffling, and efficient data loading to seamlessly handle large datasets during training. Throughout the training process, we kept the vision transformer (PanDerm) frozen to retain its pre-trained feature extraction capabilities, focusing the training on the language model and the cross-modal fusion model.

4.2.1 LOSS FUNCTION

Training utilized the CrossEntropyLoss from PyTorch to compute the difference between the predicted answers and the ground truth. The loss was calculated at each timestep of the output sequence during text generation and backpropagated to update model parameters.

4.2.2 OPTIMIZATION

The model’s parameters were optimized using the AdamW optimizer. Gradients were computed through backpropagation, which adjusts the model’s weights by passing the loss information backward through the network. The training loop included steps to clear gradients, calculate losses, and iteratively optimize the model over multiple epochs.

4.3 EVALUATION METRICS

The following metrics were used to evaluate the model’s performance:

- **BLEU Score:** Evaluates how closely the predicted answers match the ground truth by measuring the n-gram precision.
- **ROUGE Score:** Captures the similarity between the predicted answers and the ground truth by assessing the overlap of sequences, including precision, recall, and F1-score.

5 RESULTS

This section presents the evaluation of our multimodal VQA model for dermatology, focusing on quantitative and qualitative metrics. As our results are pending due to implementation challenges, we outline the experimental setup and metrics used for evaluation.

5.1 QUANTITATIVE ANALYSIS

For quantitative evaluation, we measure the model’s performance using BLEU and ROUGE scores. These metrics compare the model’s answers to the reference answers by analyzing lexical overlap and sequence quality. The evaluation spans various fusion mechanisms (e.g., CrossAttention, UNITER) and language models (e.g., BERT, GPT-2). Table 1 below summarizes the comparative analysis. Each cell in the table will record the BLEU and ROUGE scores for a specific fusion mechanism and language model pairing.

Table 1 and Table 2 below summarizes the comparative analysis. Each cell in the table will record the BLEU and ROUGE scores for a specific fusion mechanism and language model pairing.

Table 1: BLEU - Performance Comparison Across Fusion Mechanisms and Language Models (Pending Results)

Fusion Method	BERT	GPT-2	DistilBERT	T5
Cross-Attention				
UNITER				
ViLT				
FLAVA				

Table 2: ROGUE - Performance Comparison Across Fusion Mechanisms and Language Models (Pending Results)

Fusion Method	BERT	GPT-2	DistilBERT	T5
Cross-Attention				
UNITER				
ViLT				
FLAVA				

5.2 QUALITATIVE ANALYSIS

While a comprehensive qualitative analysis is beyond the scope of this paper, we plan to conduct a preliminary human evaluation. In this study:

A sample of model-generated answers will be reviewed by domain experts and lay users. Evaluators will rate the answers based on accuracy, clarity, and relevance to the provided image and question. The results from this evaluation will complement the quantitative scores to provide a holistic view of the model’s capabilities.

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